

Plot Planner Pro - User Guide

A Step-by-Step Manual for Architectural Drafting & Environmental Simulation

1. Setting Up Plot Dimensions

To start planning, configure the plot boundaries using the right sidebar panel (Irregular Plot Math Engine):

- **Boundary Sides:** Enter dimensions for Side A (Top), Side B (Right), Side C (Bottom), and Side D (Left).
- **Triangulation (Optional):** Enter Diagonal E (from top-left corner to bottom-right corner) to calculate irregular shapes using Heron's Formula. If empty, the average method will estimate the area with a warning.
- **Land Unit Selection:** Set your local Marla standard (e.g. Lahore Urban 225 sq ft or Rural 272.25 sq ft) to automatically calculate the area in Kanals and Marlas.
- **Measurement Unit:** Select Feet (ft), Meters (m), or **Yards (Karachi Yard - sq yds / yds)** to adapt the canvas and calculators to your preferred regional scale (particularly Karachi).
- **CDA Presets:** If you are in Islamabad, click the preset buttons (8 Marla or 1 Kanal) to automatically populate standard boundary dimensions.

2. Canvas Drafting and snap controls

Draw walls and partition walls using the 2D drafting board:

- Select **Draw Line** tool in the left toolbar.
- Left-click on the canvas grid to start a line, move mouse to preview length, and click again to seal. Hold **Shift** while moving to force strictly horizontal or vertical walls.
- The drafting board features smart snapping. Turn on **Grid Snapping** in the configuration settings to align wall corners automatically.

3. Regulatory Setbacks & Pin Dropping

- Select your local municipal authority from the dropdown (LDA, CDA, RUDA, SBCA, PDA, or QDA) to display mandatory setbacks.
- Standard statutory front/rear setbacks will draw as orange dashed guidelines on the canvas.
- Select a pin tool (Water, Gas, or Electric) and click anywhere on the grid to tag utility connections on your plot.

4. Running Microclimate Simulations

Use the **Geo-Energy & Microclimate** panel to simulate weather patterns and solar paths:

- Check **Solar Path & Shadows** to view the sun's trajectory. Slide the **Time of Day** slider to watch shadows cast from plot boundaries and neighbors rotate in real-time.
- Check **Animated Wind Vectors** and toggle seasons to see hot summer breezes (SW) or cool winter drafts (NE) animated across the canvas.
- Check **Sound Pollution Gradient** to see traffic noise decay zones inside the plot.

5. Estimating Construction Costs & Exporter

- Click **BOM & Costs** in the header actions. Review the estimated bricks, cement bags, sand, and tiles.
- Edit the unit rates in the pricing table to match local market rates and view the updated grand total.
- Switch roles in the header and leave team comments or municipality annotations in the approvals log.
- Click **Export Project** to save a backup JSON database file of your entire project.

6. 3D WebGL Viewport & Floor Plan Generator

Use the advanced spatial and 3D utilities to layout structures and visualize them in WebGL:

- **Procedural Floor Plan Generator:** Navigate to Step 2 (Requirements) in the right sidebar. Set floor count (Single, Double, Triple), toggle the corridor divider, check room types, input custom sizes, and click **Generate Floor Plan** to automatically populate room allocations.
- **Interactive Room Repositioning:** In Floor Plan view, left-click and drag any room block on the canvas to manually rearrange layout orientations.
- **2D Blueprint Symbols:** Visualizes wall thickness variations (9-inch outer load-bearing and 4.5-inch inner partition walls) with solid poche fills, swing-arc doors, windows, scale bars, and a North compass.
- **3D WebGL Viewport:** Click **3D View** in the header to enter WebGL viewport. Select **Orbit** to drag-rotate/zoom, or **Walk** to navigate using W/A/S/D keys. Sync with season/time sun angles, toggle roofs, and export snapshots as PNG images.
- **Multi-Format File Importer:** Drag-and-drop or click **Import File** in the left toolbar to load JSON database backups, DXF/SVG vector boundaries, STL/OBJ meshes, or PNG/JPG tracing underlays (which allow Shift-drag repositioning, opacity, and scale adjustments).
- **Drafting Undo Tool:** Click **Undo Tool** in the left sidebar or press **Ctrl + Z** to backtrack drawn walls and utility pins instantly.